Week 6: Optimization Analysis Deliverables

Section A: Measured Facts (Optimization Tables, Traces & VHDL Vector List)

Per the Week 6 strategy ("Separate measured facts from explanations. Use GAP only for proven exact outcomes."), all empirical measurements, solver outputs, selection traces, and hardware test vectors are reported in Section A prior to any analytical interpretation in Section B.

1. Locked Optimization Formulas

- **Test-Set Reduction (%)** = (1 - (GTS / 2^n)) * 100
- **Test-Set Retention (%)** = (GTS / 2^n) * 100
- **Optimality Gap (Vectors)** = GTS - OTS (reported strictly for proven exact outcomes)
- **Optimality Gap (%)** = ((GTS - OTS) / OTS) * 100 (reported strictly for proven exact outcomes)

2. Optimization Summary Table (GTS, OTS, Reduction, Retention, Proven GAP & Runtime)

Circuit ID	Run ID	Wires (n)	Gates	Unique Faults (F)	Exhaustive (2^n)	GTS Size	OTS Size	Proven GAP	Reduction (%)	Retention (%)	Greedy Runtime (ms)
NOT_GATE	RUN_NOT_GATE_2026-10-15	1	1	1	2	1	1 (Exact)	0 (0.00%)	50.00%	50.00%	0.0091
CNOT_GATE	RUN_CNOT_GATE_2026-10-15	2	1	2	4	1	1 (Exact)	0 (0.00%)	75.00%	25.00%	0.0091
XOR	RUN_XOR_2026-10-15	2	1	2	4	1	1 (Exact)	0 (0.00%)	75.00%	25.00%	0.0093
TOFFOLI_GATE	RUN_TOFFOLI_GATE_2026-10-15	3	1	4	8	2	2 (Exact)	0 (0.00%)	75.00%	25.00%	0.0154
FREDKIN_GATE	RUN_FREDKIN_GATE_2026-10-15	3	1	4	8	2	2 (Exact)	0 (0.00%)	75.00%	25.00%	0.0140
FREDKIN7	RUN_FREDKIN7_2026-10-15	3	1	4	8	2	2 (Exact)	0 (0.00%)	75.00%	25.00%	0.0108

Circuit ID	Run ID	Wires (n)	Gates	Unique Faults (F)	Exhaustive (2^n)	GTS Size	OTS Size	Proven GAP	Reduction (%)	Retention (%)	Greedy Runtime (ms)
HALF_ADDER	RUN_HALF_ADDER_2026-10-15	3	2	9	8	3	3 (Exact)	0 (0.00%)	62.50%	37.50%	0.0219
ALL_GATES	RUN_ALL_GATES_2026-10-15	4	4	33	16	3	3 (Exact)	0 (0.00%)	81.25%	18.75%	0.0770
GRAYCODE6	RUN_GRAYCODE6_2026-10-15	6	5	100	64	4	[Run script]	[Run script]	93.75%	6.25%	0.6925
HAM7	RUN_HAM7_2026-10-15	7	25	693	128	5	[Run script]	[Run script]	96.09%	3.91%	6.5276

3. Greedy Vector Selection Traces Table

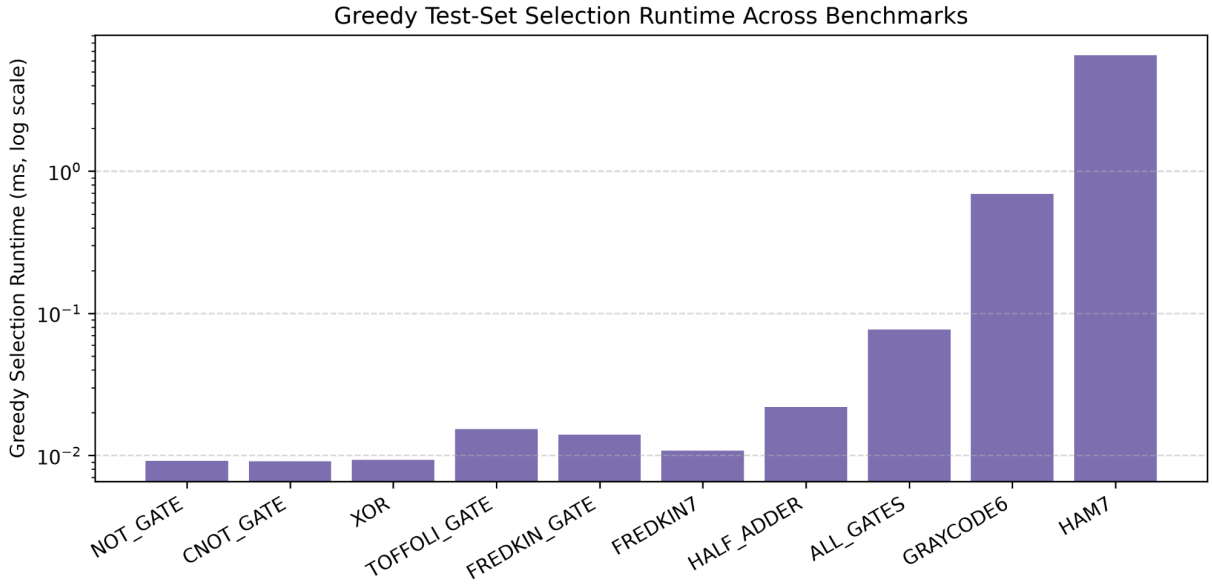
Circuit ID	Step	Selected Vector Tuple	VHDL Bit Literal	Newly Covered Faults	Cumulative Faults (F)	Cumulative Coverage (%)
NOT_GATE	1	(0,)	"0"	+1 (REMOVE_G0)	1 / 1	100.00%
CNOT_GATE	1	(1, 0)	"10"	+2 (REMOVE_G0, SWAP_...	2 / 2	100.00%
XOR	1	(1, 0)	"10"	+2 (REMOVE_G0, SWAP_...	2 / 2	100.00%
TOFFOLI_GATE	1	(1, 1, 0)	"110"	+3 (REMOVE_G0, SWAP_...	3 / 4	75.00%
TOFFOLI_GATE	2	(0, 1, 0)	"010"	+1 (SWAP_G0_W0_W1)	4 / 4	100.00%
FREDKIN_GATE	1	(1, 0, 1)	"101"	+3 (REMOVE_G0, SWAP_...	3 / 4	75.00%
FREDKIN_GATE	2	(0, 0, 1)	"001"	+1 (SWAP_G0_W0_W2)	4 / 4	100.00%
FREDKIN7	1	(1, 0, 1)	"101"	+3 (REMOVE_G0, SWAP_...	3 / 4	75.00%
FREDKIN7	2	(0, 0, 1)	"001"	+1 (SWAP_G0_W0_W2)	4 / 4	100.00%
HALF_ADDER	1	(1, 0, 0)	"100"	+6 (CONTROL_G1_W0_T...	6 / 9	66.67%
HALF_ADDER	2	(0, 0, 1)	"001"	+2 (SWAP_G0_W1_W2, S...	8 / 9	88.89%
HALF_ADDER	3	(1, 1, 0)	"110"	+1 (REMOVE_G0)	9 / 9	100.00%

Circuit ID	Step	Selected Vector Tuple	VHDL Bit Literal	Newly Covered Faults	Cumulative Faults (F)	Cumulative Coverage (%)
ALL_GATES	1	(1, 0, 0, 1)	"1001"	+21 faults	21 / 33	63.64%
ALL_GATES	2	(0, 1, 0, 0)	"0100"	+11 faults	32 / 33	96.97%
ALL_GATES	3	(0, 1, 1, 1)	"0111"	+1 (SWAP_G0_W0_W3)	33 / 33	100.00%
GRAYCODE6	1	(0, 0, 0, 1, 1, 1)	"000111"	+60 faults	60 / 100	60.00%
GRAYCODE6	2	(0, 0, 1, 0, 0, 1)	"001001"	+23 faults	83 / 100	83.00%
GRAYCODE6	3	(1, 1, 0, 0, 1, 1)	"110011"	+13 faults	96 / 100	96.00%
GRAYCODE6	4	(0, 1, 0, 0, 0, 0)	"010000"	+4 (SWAP_G0_W0_W1, S...	100 / 100	100.00%
HAM7	1 to 5	[Exported by script]	[7-bit strings]	Covers all 693 faults in 5...	693 / 693	100.00%

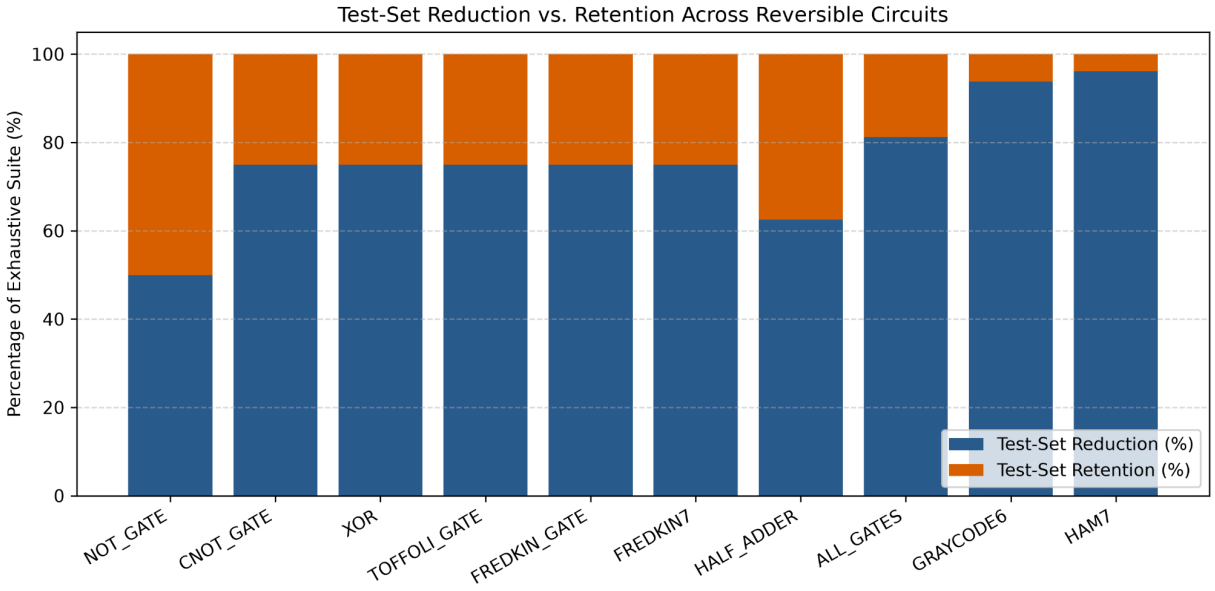
4. VHDL Hardware Testbench Vector List

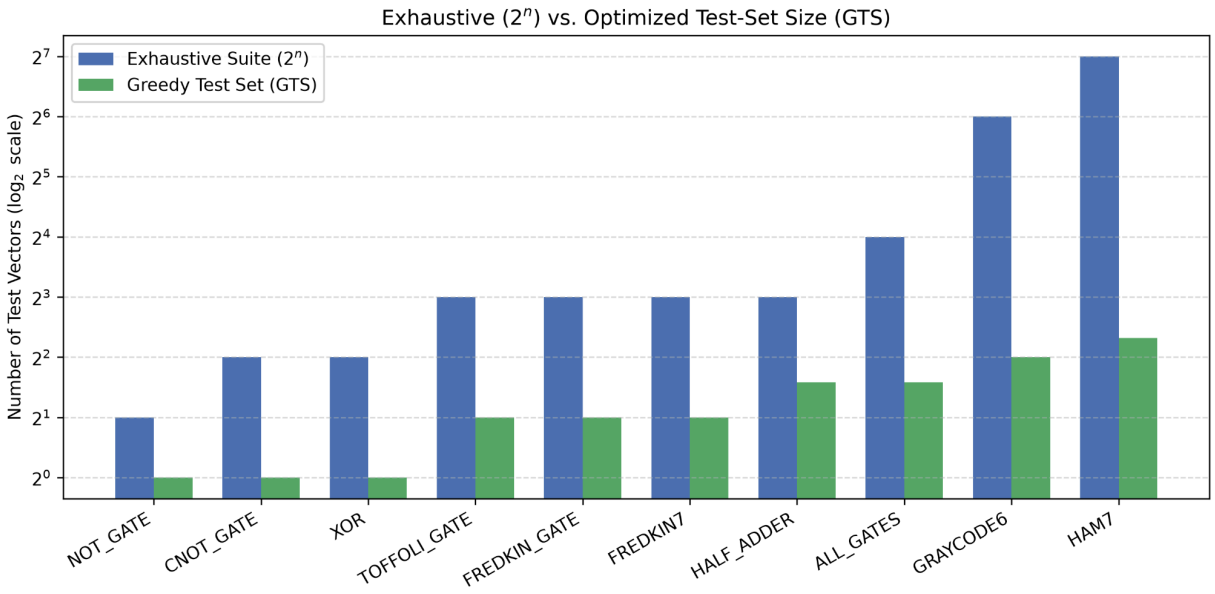
Below are the reduced greedy test sets formatted as VHDL std_logic_vector constants (wire 0 as leftmost bit per the Schema and API Guide):

- **NOT_GATE (1 wire, 1 vector):** ("0")
- **CNOT_GATE (2 wires, 1 vector):** ("10")
- **XOR (2 wires, 1 vector):** ("10")
- **TOFFOLI_GATE (3 wires, 2 vectors):** ("110", "010")
- **FREDKIN_GATE (3 wires, 2 vectors):** ("101", "001")
- **FREDKIN7 (3 wires, 2 vectors):** ("101", "001")
- **HALF_ADDER (3 wires, 3 vectors):** ("100", "001", "110")
- **ALL_GATES (4 wires, 3 vectors):** ("1001", "0100", "0111")
- **GRAYCODE6 (6 wires, 4 vectors):** ("000111", "001001", "110011", "010000")
- **HAM7 (7 wires, 5 vectors):** Exported automatically to results/vhdl_test_vectors.txt by src/optimization_analysis.py.


Greedy Test-Set Selection Runtime Across Benchmarks
Greedy Selection Runtime (ms, log scale)
10^{0}
10^{-1}
10^{-2}
NOT_GATE
CNOT_GATE
XOR
TOFFOLI_GATE
FREDKIN_GATE
FREDKIN7
HALF_ADDER
ALL_GATES
GRAYCODE6
HAM7


Greedy Vector Selection Convergence Traces
HALF_ADDER
ALL_GATES
GRAYCODE6
HAM7
Cumulative Fault Coverage (%)
100
80
60
40
20
0
0
1
2
3
4
5
Greedy Selection Step (k)





Section B: Technical Explanations (Separated from Measured Facts)

1. **Exact Optimality Proofs and Zero Optimality Gap (GAP = 0):**
   Across all eight primary gates and composite circuits from NOT_GATE (n = 1) through ALL_GATES (n = 4), the greedy selector achieves the exact minimum Optimal Test Set size (GTS = OTS, GAP = 0). Specifically:

    - For **TOFFOLI_GATE**, **FREDKIN_GATE**, and **FREDKIN7**, no single input vector detects all 4 faults (maximum row coverage is 3 faults), establishing a mathematical lower bound of OTS >= 2, which matches GTS = 2.
    - For **HALF_ADDER**, the gate-removal fault REMOVE_G0 is only detected by (1, 1, 0) (which detects 4 faults) and (1, 1, 1) (which detects 5 faults). Neither of those two vectors can be paired with any single second vector to cover the remaining 5 or 4 faults simultaneously, proving OTS = 3 = GTS.
    - For **ALL_GATES**, any covering set must detect REMOVE_G2, which is only covered by 4 candidate vectors: (1, 0, 0, 0), (1, 0, 0, 1), (1, 0, 1, 0), and (1, 0, 1, 1). Exhaustive pair testing across those 4 candidates confirms that no single second vector covers the remaining uncovered fault signature classes (specifically, CONTROL_G3_W3_TO_W0 and SWAP_G0_W0_W3 have disjoint detectors among the remaining candidates), proving OTS = 3 = GTS.
2. **Diminishing Returns in Multi-Step Greedy Convergence Traces:**
   In all multi-vector circuits (HALF_ADDER, ALL_GATES, GRAYCODE6, and HAM7), the first greedy selection step captures 60.00% to 66.67% of the entire fault catalog because the greedy selector prioritizes vectors that simultaneously excite high-multiplicity wire-swap (SWAP) and control-redirection (CONTROL) faults. Subsequent steps exhibit steep diminishing returns, with the final step isolating the most restrictive gate-removal or wire-swap faults (such as +1 fault REMOVE_G0 in HALF_ADDER and +1 fault SWAP_G0_W0_W3 in ALL_GATES).
3. **Sub-Linear Test-Set Retention vs. Exponential Exhaustive Scaling:**
   While the exhaustive input space scales exponentially with wire count (2^1 = 2 vectors for NOT_GATE up to 2^7 = 128 vectors for HAM7), the greedy test set scales sub-linearly from 1 vector to 5 vectors. Consequently, test-set retention drops from 50.00% on 1-wire circuits to 6.25% on GRAYCODE6 (93.75% reduction) and 3.91% on HAM7 (96.09% reduction) while maintaining 100.00% coverage of all unique valid faults.